MACHINE LEARNING – LEARNING PLAN

Phase	Topics	Duration
Phase 1: Programming Fundamentals	Learn Python programming, variables, data types, operators, conditional statements, loops, functions, object-oriented programming and basic problem-solving.	Weeks 1–4
Phase 2: Mathematics and Statistics	Learn linear algebra, matrices, vectors, probability, statistics, mean, median, mode, standard deviation, variance and basic calculus required for Machine Learning.	Weeks 5–7
Phase 3: Data Analysis and Preprocessing	Learn NumPy, Pandas, Matplotlib and Seaborn. Understand data collection, data cleaning, handling missing values, data preprocessing, exploratory data analysis (EDA) and data visualization.	Weeks 8–10
Phase 4: Machine Learning Fundamentals	Learn supervised and unsupervised learning, regression, classification, clustering, decision trees, random forests, KNN, SVM, model training, testing and evaluation.	Weeks 11–14
Phase 5: Deep Learning and Advanced Technologies	Learn Artificial Neural Networks (ANN), CNN, RNN, LSTM, backpropagation, TensorFlow, Keras, PyTorch, Computer Vision, NLP and basic Generative AI concepts.	Weeks 15–19
Phase 6: Projects and Career Preparation	Build practical Machine Learning and Deep Learning projects, learn Git and GitHub, practise real-world datasets, explore model deployment, prepare a resume, build a portfolio and practise technical interviews.	Weeks 20–24